How to use this script: each slide has a one-sentence key message (say this even if you skip everything else), plain-English talking points with everyday analogies, and a short ‘If asked’ section with the technical detail.
Running analogy: volatility is the weather of the stock market. GARCH is the weather forecast. MCMC and VI are two different ways of working out how much to trust that forecast.

Slide 1: Title – Bayesian Volatility at Scale

Key message: We test whether a fast machine-learning shortcut can replace a slow, trusted statistical method for forecasting market risk.

- Today I’m presenting ‘Bayesian Volatility at Scale’.
- My project is based on a 2023 research paper by Magris and Iosifidis. I re-ran their Microsoft example from scratch in Python, checked my numbers against theirs, and then extended it.

Slide 2: The Project in Plain English

Key message: Volatility is how bumpy a stock’s ride is, and we compared two ways of forecasting it with honest error bars.

- What is volatility? It is how big the daily price swings are. Like the weather, stormy days tend to follow stormy days.
- Why does it matter? Banks and investors use volatility forecasts to set risk limits, price options and decide how much to invest.
- What did I compare? A trusted but slower method called MCMC against a machine-learning shortcut called Variational Inference, or VI.
- What did I find? The two give the same forecasts. The simple version of the shortcut is over-confident, the full version is not, and for a model this small the shortcut isn’t actually faster.
- Analogy: MCMC explores a landscape on foot; VI flies over it and draws a simple map. The question is whether the map is good enough to navigate by.

Slide 3: MCMC – the Gold Standard

Key message: The traditional method is reliable but works by trial and error; VI turns the problem into ‘turn the dial until the score stops improving’.

- On the left is MCMC. It explores possible answers by taking thousands of small random steps, like searching a dark room by feel. It gets there, but it needs careful tuning and checking.
- On the right is Variational Inference. Instead of wandering, it optimises: it adjusts a simple curve until it fits as well as possible, the same way machine-learning models are trained.
- I’ll be upfront: for our small model, MCMC turned out not to be slow. VI’s advantages are elsewhere, and I’ll show them.

If asked (technical detail):

- MCMC here is a Random-Walk Metropolis sampler. VI maximises the ELBO with momentum or Adam updates.

Slide 4: Recasting the Problem as Optimisation

Key message: The exact answer is too complicated to calculate, so VI finds the closest simple bell curve instead.

- Panel 1: the true answer, the ‘posterior’, is a complicated shape we can’t compute exactly.
- Panel 2: we choose a family of simple shapes, here bell curves.
- Panel 3: we slide and stretch the bell curve until it overlaps the true shape as closely as possible. That is an optimisation problem, which computers are very good at.

If asked (technical detail):

- The ‘closeness’ measure is the Kullback–Leibler (KL) divergence; the variational family is Gaussian, $N(\mu, \Sigma)$.

Slide 5: The ELBO – a Fit Score

Key message: The ELBO is a single score that rewards fitting the data and penalises straying far from sensible starting beliefs.

- Think of the ELBO as a fit score: the higher, the better.
- The left part rewards the model for explaining the data well. The right part is a penalty for drifting too far from our starting beliefs, which stops the model from over-fitting.
- Because the penalty is already built in, we don't add any extra penalty on top. That was a mistake in my first draft, and I'll come back to it.

If asked (technical detail):

- ELBO = $E_q[\log p(y|\theta)] - KL(q(\theta) \| p(\theta))$. The expectation is estimated by Monte Carlo: 50 draws per step for BBVI, 10 for full-covariance VI.

Slide 6: Guardrails on the Model

Key message: The model has three dials, and we give them built-in guardrails so it can never produce an impossible forecast.

- GARCH has three dials: a baseline level of turbulence (ω), how strongly the market reacts to yesterday's shock (α), and how long turbulence is remembered (β).
- If the dials go out of range, the forecast can become negative or explode. Following the paper, we pass the dials through simple functions so that every possible setting is automatically safe, like a thermostat that physically can't be set above a limit.
- This replaced my first draft, which used an ad-hoc 'if too big, shrink it' rule.

If asked (technical detail):

- Transforms (paper, Proposition 2): $\omega = \exp(\theta\omega)$, $\alpha = \text{sigmoid}(\theta\alpha)$, $\beta = \text{sigmoid}(\theta\beta)(1 - \alpha)$, which guarantees $\alpha + \beta < 1$ (stationarity) for every θ.

Slide 7: The Training Loop

Key message: VI learns by repeating guess, score, nudge until the score stops improving.

- The computer guesses some dial settings, scores how well they explain the data, nudges the guess in a better direction, and repeats about 1,500 times.
- I built two versions. The paper's version, BBVI, only needs the scores, but it assumes the three dials are unrelated. The full version, in PyTorch, lets the dials move together, which turns out to matter.

If asked (technical detail):

- BBVI: score-function (log-derivative) gradient, diagonal covariance, S = 50, learning rate 0.005, momentum 0.4, 2,500 iterations (the paper's settings).
- Full-covariance VI: reparameterisation $\theta = \mu + L\varepsilon$, gradients by automatic differentiation, S = 10, Adam with learning rate 0.01.

Slide 8: The Data

Key message: We used exactly the paper's data, Microsoft from 2018 to mid-2023, so we can check our answers against theirs.

- The data are Microsoft's daily share prices from January 2018 to June 2023, 1,383 trading days, the same as the paper.
- I checked for gaps, duplicates and errors and found none. Then I turned prices into daily percentage changes.
- The models learn from the first 75% of days. The last 25%, which include the 2022 sell-off, are kept as an unseen 'exam'.

If asked (technical detail):

- Returns are 100 × the change in log price. Train: Jan 2018 – 11 Feb 2022 (1,036 days); test: 346 days.

Slide 9: What the Data Look Like

Key message: Turbulence comes in waves, and crashes are more common than a 'normal' bell curve suggests.

- First, extreme days, both crashes and spikes, happen far more often than a bell curve predicts. These are called fat tails.
- Second, stormy days follow stormy days. Today's swings help predict tomorrow's.
- Third, the average daily gain is tiny compared with the daily swings, so we focus on modelling the swings.

- GARCH is built to capture the waves; our Student-t extension handles the crashes.

If asked (technical detail):

- Excess kurtosis 9.5; Jarque-Bera $p < 0.001$; Ljung-Box(10) on squared returns 890; ARCH-LM(10) 393, all $p < 0.001$.

Slide 10: We Match the Published Paper

Key message: Our numbers match the paper's to within about 0.02, so the code is doing the right thing.

- The teal columns are my estimates; the gold columns are the paper's. They line up closely for every method.
- The dials tell a story: $\beta \approx 0.77$ means turbulence has a long memory. A volatility shock takes about 15 trading days to fade by half.

If asked (technical detail):

- QML: $\omega = 0.160$, $\alpha = 0.186$, $\beta = 0.769$ versus 0.15, 0.19, 0.77 in the paper. Test NLL 764.6 versus 761.4.

Slide 11: How Sure Are We?

Key message: All methods agree on the best guess, but the simple VI is over-confident; the full VI gets the uncertainty right.

- Each curve shows the range of plausible values for a dial. A wider curve means less certainty.
- The full VI (dashed) lies right on top of the trusted MCMC (grey).
- The simple VI (gold) is centred correctly but far too narrow. It claims to be about twice as sure as it should be, like a weather forecaster who never says 'maybe'.
- The reason is that it ignores how the dials move together. The paper finds exactly the same thing.

If asked (technical detail):

- Posterior SD of ω: MCMC 0.049, full VI 0.050, BBVI 0.022. Full VI also reaches a higher ELBO (-1945.3 vs -1946.2).

Slide 12: Forecasts with a Range

Key message: Bayesian methods give a forecast with a 'likely range', like a weather forecast that says 18–22 degrees.

- The line is the forecast turbulence for each day; the shaded band is its likely range.
- The model picks up the late-2018 sell-off, the COVID-19 crash in March 2020, when annualised volatility peaked above 150%, and the 2022 sell-off, which it never saw while learning.
- A classic single-number method can't give you this range. The full VI's range matches MCMC's; the simple VI's is about 10% too narrow.

If asked (technical detail):

- Average 90% band width in the test period: MCMC 3.5, full VI 3.5, BBVI 3.1 volatility points.

Slide 13: Accuracy and Cost

Key message: All methods forecast equally well, but for this small model the shortcut is not faster.

- On the left: on the unseen 'exam' data, every method's forecast errors are within 1% of each other. VI gives up essentially nothing in accuracy.
- On the right, the honest part: the trusted MCMC took about a third of a second, while VI took 1 to 16 seconds. So for a model with only three dials, VI is not a speed win, and the paper never claimed it was.
- VI's real advantages are that it is easy to monitor, needs no hand-tuning per stock, and scales to much bigger models where MCMC struggles.

If asked (technical detail):

- Test-set gaps versus QML: NLL +0.15–0.19%, QLIK +0.26–0.32%. Likelihood evaluations: MCMC 30,000, BBVI 125,000, full VI 15,000.

Slide 14: Allowing for Crash Days

Key message: A model that expects the occasional crash day fits real markets clearly better.

- The standard model assumes a bell curve, so real crash days surprise it. The left chart shows those surprises at the ends.
- The Student-t version expects occasional extreme days, and on the right chart the points line up.
- It scores better both on the fit score and on the unseen data. Interestingly, the forecast line barely changes: what improves is how well the model prepares for extreme days, which is what matters for risk management.
- My first draft also added an extra 'MAP penalty'. That counted our starting beliefs twice, so I removed it to keep the comparison fair.

If asked (technical detail):

- $\nu = 2 + \exp(\theta_\nu)$; posterior mean $\nu \approx 7.4$ (90% interval 5.2–10.2). ELBO +10.7 and test log-likelihood +11.8 versus Gaussian.

Slide 15: Is Microsoft a Fluke?

Key message: Repeating the test on 28 large US companies gives the same answer.

- To make sure Microsoft wasn't a special case, I ran the same comparison on the Dow Jones companies, with no re-tuning for any of them.
- Two were dropped because their price history was incomplete: Dow Inc. only listed in 2019, and Walgreens went private.
- Across the other 28, the methods' forecasts differ by about 0.1–0.3% on average, the same picture as the paper's study of 488 stocks.

If asked (technical detail):

- Mean % deviation from QML on test NLL: MCMC +0.12, BBVI +0.14; QML has the lowest training NLL for 100% of stocks, as expected.

Slide 16: Summary

Key message: VI is an accurate shortcut for Bayesian volatility forecasting, provided you use the full version.

- One: our results match the published paper.
- Two: VI forecasts are as accurate as the trusted methods, and the full version also gets the uncertainty right.
- Three: allowing for crash days improves the model. For a small model like this, VI isn't faster; its pay-off comes with bigger models.
- Thank you. I'm happy to take questions.

If asked (technical detail):

- Next steps: other GARCH variants (GJR, EGARCH, FIGARCH), the paper's natural-gradient optimisers, and intraday data for a sharper measure of true volatility.

Slide 17: Jargon Buster (backup)

Key message: Use this slide if anyone asks what a term means.

- No need to present this slide; keep it as a reference during questions.

Question 1: Why switch to Student-t, and what happened to the MAP regularisation?

Question: "Why did you move from a Gaussian baseline to Student-t innovations, and why was the MAP penalty removed?"

Answer:

- In plain terms: real markets have more crash days than a bell curve allows, and the Student-t model expects them. It scores better on both the fit score and the unseen data, by more than 10 points each.
- The MAP penalty in my first draft added an extra penalty on top of one that was already in the objective, so our starting beliefs were counted twice and the scores were no longer comparable. Removing it makes the comparison fair.

Question 2: If VI is not faster here, why use it?

Question: “Your MCMC was faster than VI. Doesn’t that undermine the case for VI?”

Answer:

- For a model with only three dials, nothing is slow, and a well-tuned MCMC is very efficient.
- But my MCMC needed hand-tuning, while VI used the same settings for all 28 stocks with no tuning at all.
- VI’s real benefit appears in much bigger models with hundreds or thousands of parameters, where MCMC slows to a crawl. The key finding, that VI is as accurate as MCMC, still holds.

Question 3: Why is the simple (diagonal) VI over-confident?

Question: “Why did the diagonal BBVI underestimate the uncertainty?”

Answer:

- The dials are linked: a market that reacts strongly to shocks tends to have a shorter memory, and vice versa.
- The simple VI assumes the dials are unrelated, so it misses that link and ends up too narrow. In other words, it is too sure of itself.
- The full version captures the link and matches MCMC, which is exactly what the paper reports for its full-covariance methods.